


# *Standard Operating Procedure KIA* CNC Mill

## KIA CNC Mill

The **KIA** is a 3-axis CNC vertical machining center featuring a 80**00-RPM** inline direct-drive spindle and rapids of **600 IPM** (inch per minute). This machine is an automated, high-precision tool. Unlike manual equipment, it can initiate rapid motion at any time during a program cycle.

## Potential Hazards

- Be aware of the following potential hazards when working with band saws:
    - Mechanical: Rapid axis movement and the Side-Mount Tool Changer (SMTC) present severe pinch and crush points. Never place hands between the tool changer and the spindle head.
    - Ejection: Improperly clamped parts or oversized stock can be ejected with deadly force, potentially puncturing the safety enclosure.
    - Electrical: High voltage 220V is present in the control cabinet. The cabinet must remain locked and accessible only to qualified personnel.
    - Chemical: CNC coolants can be corrosive to eyes and skin; avoid ingesting or inhaling mist.

## Before Use

- **Balanced Tooling**: DO not operate above 6000 rpm
- **Clearance Check**: Ensure the work zone is clear of loose tools and the Z-axis height is verified for all tool lengths % part fixturing.
- Tool changer swings the tool down and counterclockwise expect to lose 3 inches of clearance during a tool change.

## During Use

- **First-Run Safety**: Maintain a hand on the **Emergency Stop** button during the first run of any new program. Use 5% rapid overrides until tool paths are verified. Low Initial feed rate is also recommended to verify correct Speed and feed rate.
- **Enclosure**: Keep doors closed and guards in place during all automatic operations.
- **Work holding**: Ensure pressure is set correctly and the workpiece is securely clamped. Do not machine parts larger than the work holding capacity.




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## After Use

- **Chip Clearing**: brushes only when all machine operation has stopped & the machine is powered down
- **Internal Cleanup**: Do not use compressed air for cleaning; it forces fine particulates into precision bearings and under way-covers.
- **Power Down**: Home the machine and depress red power off button on pendant, before engaging the main circuit breaker.

## Emergency Equipment and Procedures

- **E-Stop**: The large red circular button on the pendant immediately stops all motion. In a power loss event, be aware that the spindle will take significantly longer to coast to a stop.
- **Incident Reporting**: Immediately report any tool crashes, enclosure damage, or near misses to the Laboratory Project Supervisor via phone.
- Know the location of the nearest emergency equipment and items, such as the emergency stop/shut-off for the specific machine, eye wash, first aid kit, fire extinguishers, fire alarm pull stations, and emergency exits.



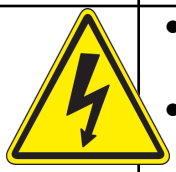

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## Appendix B - Job Hazard Analysis

| Work Duties | Potential Hazards | Controls |
|---|---|---|
| Operating the Machine | **Flying Debris**  | • Safety glasses are required while o<br>• Machine mounted guards must b an appropriate manner |
| Operating the Machine | **Pinch, Crush and Entanglement Hazards**  | • Use caution while handling heavy devices |
| Operating the Machine | **Entanglement, Lacerations and Flying Debris Hazards** | • During Tool Changes, machine mu<br>• Operator must be trained in th use of the brake to prevent startup |




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| | | • Never reach towards the operation while machine is runn |
| Operating the Machine | **Fire**  | • Operators are trained to shut d machine in accordance with outlined in the machine-specific SOP<br>• All combustibles must be clear of th metal shavings, combustible dust debris must be cleaned up appropria<br>• Class D extinguishers must be acces times |
| Operating the Machine | **Slips, Trips and Falls**  | • Proper housekeeping must be m at all times, including cleaning immediately |
| Machine Service | **Electrocution/Accidental Start up**  | • Always follow Lock out / Tag out control of hazardous energy<br>• Equipment needing serviced shall out immediately by authorized per<br>• Service should be completed in the manufacturer requirements b and authorized personnel or a cont |
| Post-Operation Housekeeping | **Lacerations or Foreign Body in Eyes**  | • When cleaning up safety glasses a must be worn at all times<br>• NEVER attempt cleaning up swarf/cl machine is running<br>• Gloves should be used with cautio machinery, as they may pre entanglement hazard |